$$PV_t = FVe^{-r\,t}. \quad (4)$$

Fixed-Income Instruments and the Time Value of Money

Fixed-income instruments are debt instruments, such as a bond or a loan, that represent contracts under which an issuer borrows money from an investor in exchange for a promise of future repayment. The discount rate for fixed-income instruments is an interest rate, and the rate of return on a bond or loan is often referred to as its yield-to-maturity (YTM).

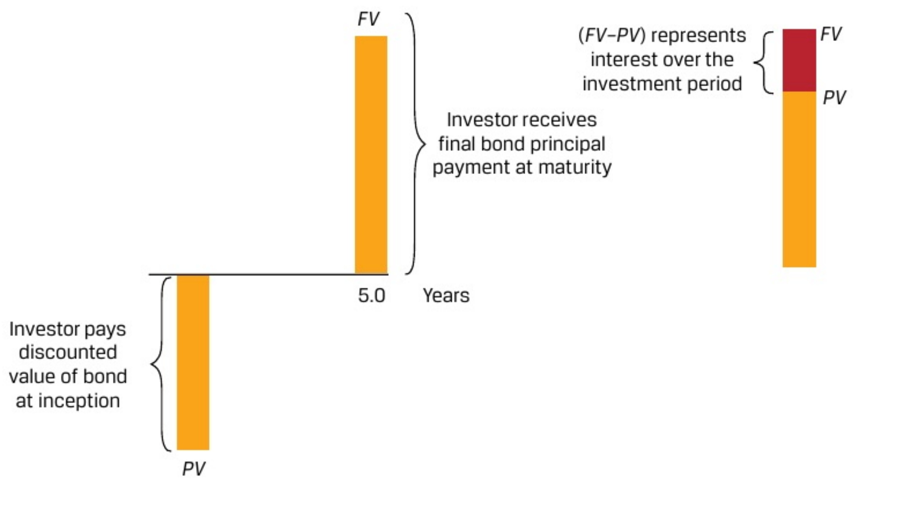
Cash flows associated with fixed-income instruments usually follow one of three general patterns:

- Discount: An investor pays an initial price (*PV*) for a bond or loan and receives a single principal cash flow (*FV*) at maturity. The difference (*FV* – *PV*) represents the interest earned over the life of the instrument.
- Periodic Interest: An investor pays an initial price (*PV*) for a bond or loan and receives interest cash flows (*PMT*) at pre-determined intervals over the life of the instrument, with the final interest payment and the principal (*FV*) paid at maturity.
- Level Payments: An investor pays an initial price (*PV*) and receives uniform cash flows at pre-determined intervals (*A*) through maturity which represent both interest and principal repayment.

Discount Instruments

The discount cash flow pattern is shown in Exhibit 1:

Exhibit 1: Discount Bond Cash Flows



The present value (*PV*) calculation for a discount bond with principal (*FV*) paid at time t with a market discount rate of r per period is:

$$PV(\text{Discount Bond}) = FV_t\,/\,(1 + r)^t. \quad (5)$$